LILY ZHANG

San Francisco Bay Area | lilyzhng.ai@gmail.com | linkedin.com/in/lilyzhng | lilyzh.ng | github.com/lilyzhng | @lily_gpupoor

STATEMENT

Applied research lead with 7 years managing cross-functional AI training data. I have led global ML engineering teams across 4 countries, shifted an entire org from traditional ML pipelines to agentic workflows, and taken research from prototype to production. My focus is post-training, AI training data including RL environments, and evaluation. 8 publications at CVPR, NeurIPS, ICCV.

HIGHLIGHTS

- **Built RL Environments and Eval Benchmarks:** Built SuperGeneral, compositional tool environments for long-horizon agents on APEX benchmark; Designed verifiable failure taxonomy for RL reward shaping (tested on Terminal Bench & τ^2-bench). Built LosslessBench evaluating inference acceleration impact on model behaviors.
- **Large-Scale Training & Resource Management:** Owned compute allocation for 1000+ GPUs across competing teams, designed a dynamic priority scheduling system for cluster sharing. Built SofaGenius, an automated post-training platform, productionized for internal ML experiments.
- **0-to-1 Program Delivery:** Decomposed a fuzzy research goal into a phased roadmap with milestones, owners, and review checkpoints. Stood up a scout team across perception, mapping, ops, and labeling. AI training data research shipped in Ford F-150 Lightning and Super Duty; auto-labeling pipeline delivered $1M+ savings.
- **Program Leadership & Org Building:** Ran a global team of 10+ across 4 countries (US, India, Germany, Israel). Turned ambiguous customer data contracts into scoped deliverables with named owners, milestones, and a recurring PI-planning that aligned perception, ops, and labeling. At Latitude AI, drove the org-wide shift from traditional ML to agentic workflows. Skills/Tooling I built is now used by 76+ engineers.
- **Alignment Research & Governance Tooling:** C-Guard: A Constitution-Grid Instrument for Data-Efficient RL Alignment (under review, COLM 2026 Efficient Reasoning). Turns the helpfulness/harmlessness tradeoff into measured, gated moves on a constitution grid. As research advisor at Alibaba Cloud, led Eureka from research to launching, monitored operational health, +16% demand fulfillment, −33% compute migration, 91% ops adoption. Post @ Neurips 25 VLM4RWD, Oral @ DASFAA 2026.

LEADERSHIP & SERVICE

- **NVIDIA GTC Conference.** Keynote on synthetic data generation using AI @ GTC 23. Panelist at AI/ML infra @ GTC 26
- **Industry Advisory Board:** Board Advisor for Texas A&M University, University of Delware.
- **Research Advisor, AliCloud (2025)** Led the research direction for Eureka, AI infra agent post-training.
- **Program Committee:** NeurIPS, CVPR, ICRA, IROS, ICML, ICLR, ECCV, KDD, ICCV

EXPERIENCE

Technical Lead, AI & ML | Latitude AI Feb 2024 - Present

- Managed compute allocation for 1000+ GPUs over 2 months to train 50000 models. Designed dynamic priority system with Dagster for cluster sharing across teams.
- Championed VLM/LLM adoption across the org. Built Alfa Agent for data selection and auto QA; adopted org-wide with 76+ users.
- Formed a cross-functional scout team spanning perception, mapping, operations, and labeling; defined and phased the applied research roadmap for ML-based satellite-imagery segmentation, setting milestones for each phase.
- Designed agentic QA platform (Aqua), focusing on bring your own tool (BYO) for cost-effective labeling, automatic triage, and rework.

Technical Lead, AI & ML | Ford Greenfield Labs Jun 2022 - Jan 2024

- Influenced global team without authority, working with data-generation team of 10+ engineers across US, India, Germany, and Israel. Drove cross-org alignment and PI planning, turning customer data contracts into regular deliverables across perception workstreams.
- Transformed the research-scientist team into a applied ML engineering group: established production coding standards and mentored researchers into ML engineers with a product mindset and due diligence.
- Proposed and built auto-labeling pipeline replacing expensive manual annotation. Delivered 15K 3D annotations and $1M+ in savings. Pipeline covers both perception labels and VLM-generated scene captions.
- Partnered with external groups, including Voxel51, Scale AI, Google, and NVIDIA on data and ML tooling.

Sr. Research Scientist, AI & ML | Ford Greenfield Labs Jun 2019 - May 2022

- Led full ML lifecycle from research prototype to production: synthetic data augmentation shipped in F-150 Lightning and Super Duty (Pro Trailer Hitch Assist). Bridged research and product teams.
- Pioneered AI-based 3D scene relighting, enabling reliable perception with cost-effective sensor configurations.

PRODUCTS & OPEN SOURCE

- **SuperGeneral (2026)** Compositional tool environments for long-horizon agents on APEX benchmark. Tested 4 frontier models on tool use, composition, and creation.
- **RLVR Environment Design from Failure Taxonomy (2026)** What Can't Be Measured Can't Be Solved. Designed a verifiable failure taxonomy for RL reward shaping, benchmarked on Terminal Bench & τ^2-bench.
- **GeniusTeam (2026)** Multi-agent orchestration. 4 specialized agents coordinate autonomously with file-based handoffs and worktree isolation.
- **SofaGenius Post-Train (2026)** Automated post-training platform. GRPO on 80B MoE models, cost-visible GPU launches on Modal, 7 anomaly detectors for model convergence.
- **Frontend Slides (2026)** Claude Code skill for HTML presentations. Reached tens of thousands of users, 25K+ github stars.

SELECTED PUBLICATIONS

- **Eureka** (Oral DASFAA 2026, NeurIPS-W 2025) Agent that writes forecasting features as code.
- **LiHi-GS** (RAL/CVPR-W 2025) LiDAR-supervised Gaussian splatting for highway driving scene reconstruction.
- **SIMBAR** (CVPR 2022) Single image relighting for data augmentation. Diversifying ML training data.
- 7 granted US/CN/DE patents on synthetic data generation, augmentation, and curation.

SKILLS

- **Technical Program Management:** scoping ambiguous problems, roadmap and milestone planning, cross-functional coordination.
- **LLM Post-Training:** GRPO, DPO, LoRA, Reward Modeling, Chain-of-Thought, SFT.
- **Evaluation & Benchmarking:** Benchmark Design, Rubric Design, Automated Evaluators, Failure Analysis, Scoring Frameworks
- **Agent Infrastructure:** Multi-Agent Orchestration, Tool Use, Compositional Environments, MCP, Long-Horizon Agents
- **ML at Scale:** Distributed Training, MoE Architecture, Perception Data Flywheels, Modal, Synthetic Data Pipelines
- **Programming:** Python, C++, PyTorch, TensorFlow, Transformers, SQL, Cloud Platforms

EDUCATION

- Stanford University, AI Graduate Program (2022-2024)
- Penn State University, B.S. Computer Science (2013-2017)